

Qualification Pack



Associate-Clinical Research Management (Pharma, Biologics and Medical devices)

Electives: Site management/ Study Monitoring/ Data management /
Biostatistics

QP Code: LFS/Q3501

Version: 3.0

NSQF Level: 5.5



Qualification Pack

Life Sciences Sector Skill Development Council || # 14, Rear 2nd Floor, Palam Marg, Vasant Vihar
New Delhi-110057 || email:SHIVI.CHAUDHARY@LSSSDC.IN



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LFS/Q3501: Associate-Clinical Research Management (Pharma, Biologics and Medical devices)

Brief Job Description

Associate-Clinical Research Management (Pharma, Biologics and Medical devices Facility) supports clinical trial activities, carries out reporting and documentation for monitoring of research activities to ensure regulatory compliance and current Good Clinical Practices (GLP) as per ICH and coordinates with site staff members, investigators, Site Management Organization and Sponsor. The role holder is expected to assist in the Biostatistics analysis of clinical trial data with the help of artificial intelligence.

Personal Attributes

The individual should have excellent communication skills and analytical skills. The person should possess good critical thinking, attention to detail, and decision-making skills. The role holder should be proactive in planning and organizing skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [LFS/N3507: Introduction to life sciences industry and basics of Clinical trial occupation](#)
2. [LFS/N0119: Ensure environment sustainability and sensitivity towards all genders and people with disability](#)
3. [LFS/N3506: Carry out management activities related for clinical trial](#)
4. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Electives(mandatory to select at least one):

Elective 1: Site management

This elective is about Associate-Clinical Research Management performing site management activities.

1. [LFS/N3503: Perform site management activities at clinical research site](#)
2. [LFS/N3504: Carry out reporting and documentation for site coordination activities as per regulatory standards](#)

Elective 2: Study Monitoring

This elective is about Associate-Clinical Research Management performing clinical study monitoring activities.



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1. [LFS/N3501: Monitor the clinical trial site to ensure that ICH GCP guidelines, study protocol and applicable regulations are followed](#)
2. [LFS/N3502: Carry out reporting and documentation for site monitoring activities as per regulatory standards](#)

Elective 3: Data management

This elective is about Associate-Clinical Research Management providing support for clinical data management activities.

1. [LFS/N3505: Provide support for clinical data management activities](#)

Elective 4: Biostatistics

This elective is about Associate-Clinical Research Management providing support for Biostatistics analysis

1. [LFS/N3508: Provide support in data through Biostatistical analysis](#)
2. [LFS/N3509: Provide support in PK & PD through Biostatistics analysis](#)
3. [LFS/N3510: Provide algorithm support for Biostatistics activities](#)
4. [LFS/N3511: Apply AI/ML Models in biostatistical analysis](#)

Qualification Pack (QP) Parameters

Sector	Life Sciences
Sub-Sector	Pharmaceutical, Bio Pharmaceutical, Medical Devices and In Vitro Diagnostic (IVD), Biotechnology, Contract Research
Occupation	Clinical Trials, Bio Availability-Bio Equivalence
Country	India
NSQF Level	5.5
Credits	58
Aligned to NCO/ISCO/ISIC Code	NCO 2015/3512.0601

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Minimum Educational Qualification & Experience	Completed 3 year UG degree (Biology, Nursing, Medical Lab Technician, Life Sciences, Biotechnology) OR Completed 3 year UG degree program after 12th (B.E/ B. tech. Biotech) OR Completed 3 year UG degree program after 12th (B. Pharma) OR Medical Graduate (BAMS/ BDS/ BUMS/ BHMS/ MBBS)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/04/2028
NSQC Approval Date	08/04/2025
Version	3.0
Reference code on NQR	QG-5.5-LS-00251-2025-V2-LSSSDC
NQR Version	2.0



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LFS/N3507: Introduction to life sciences industry and basics of Clinical trial occupation

Description

This NOS is about an Associate-Clinical Research Management Individual Introduction to life sciences industry and basics of Clinical trial occupation

Scope

The scope covers the following :

- Life sciences industry and clinical trial occupation

Elements and Performance Criteria

Life sciences industry and clinical trial occupation

To be competent, the user/individual on the job must be able to:

- PC1.** Identifies major players, key products, and innovations in the life sciences industry.
- PC2.** Explain the key regulations and legislations governing clinical research in India (e.g., Schedule Y, ICH-GCP) and compares them with global standards.
- PC3.** Follow key regulations and legislations governing clinical research
- PC4.** Identify the workflow and key responsibilities in clinical trial roles
- PC5.** Explain the importance subject/patient confidentiality and consent.
- PC6.** Explains the impact of errors or negligence by unskilled personnel on clinical trials and public health.
- PC7.** Demonstrates the ability to identify gaps in compliance in a clinical trial

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles of ICH-GCP, Indian GCP and ICMR guidelines, Good Documentation Practices and Good Laboratory Practices
- KU2.** SOPs and organizational policies about communication, code of conduct
- KU3.** clinical research team reporting structure
- KU4.** correct method for carrying out corrective actions outlined for trial-related problems
- KU5.** sponsors and CRO roles and responsibilities
- KU6.** site roles and responsibilities
- KU7.** working in cross functional, cross geographical and cross -cultural teams
- KU8.** information security and confidentiality policy
- KU9.** clinical trial-related regulations and compliances
- KU10.** sample handling procedures
- KU11.** principles of trial protocol



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use written communication skills to accurately record the required details in a given format as per SOP and guidelines in the English language
- GS2.** use reading and comprehension skills of various coding systems to read instructions, guidelines, procedures, rules, and signage
- GS3.** use listening skills to interpret the instructions and procedures to be followed
- GS4.** use verbal communication skills to interact with supervisor, teammates, and cross-functional teams for coordination and to communicate confidential and sensitive information discretely to the authorized person
- GS5.** use team-building skills while dealing with teammates and while managing the difficult/stressful or emotional situations at work
- GS6.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS7.** apply planning and organizing skills to plan and organize tools and material required to fulfil documentation related requirements
- GS8.** apply critical thinking skills to analyze and identify what and when to report an issue/concern to the supervisor/ QA team / any other stakeholder
- GS9.** apply customer-centricity while generating and securing documents
- GS10.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS11.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Life sciences industry and clinical trial occupation</i>	60	-	20	20
PC1. Identifies major players, key products, and innovations in the life sciences industry.	-	-	-	-
PC2. Explain the key regulations and legislations governing clinical research in India (e.g., Schedule Y, ICH-GCP) and compares them with global standards.	-	-	-	-
PC3. Follow key regulations and legislations governing clinical research	-	-	-	-
PC4. Identify the workflow and key responsibilities in clinical trial roles	-	-	-	-
PC5. Explain the importance subject/patient confidentiality and consent.	-	-	-	-
PC6. Explains the impact of errors or negligence by unskilled personnel on clinical trials and public health.	-	-	-	-
PC7. Demonstrates the ability to identify gaps in compliance in a clinical trial	-	-	-	-
NOS Total	60	-	20	20



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3507
NOS Name	Introduction to life sciences industry and basics of Clinical trial occupation
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials
NSQF Level	5.5
Credits	1.00
Version	1.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025



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LFS/N0119: Ensure environment sustainability and sensitivity towards all genders and people with disability

Description

This NOS unit is about role holder ensuring environment sustainability and sensitivity towards gender, and people with disabilities at the workplace

Scope

The scope covers the following :

- Environment sustainability
- Sensitivity towards all genders and people with disability

Elements and Performance Criteria

Environment sustainability

To be competent, the user/individual on the job must be able to:

- PC1.** ensure energy conservation by switching off the machine and equipment post operations
- PC2.** identify ways to optimize the usage of electricity/energy in various tasks/activities/processes
- PC3.** ensure energy conservation by optimizing the machine/ equipment performance
- PC4.** identify recyclable and non-recyclable, and hazardous waste generated
- PC5.** segregate waste into different categories to achieve minimum pollution of land and water
- PC6.** check for water leakage in plant/ work area and take corrective actions

Sensitivity towards all genders and people with disability

To be competent, the user/individual on the job must be able to:

- PC7.** respect all genders, religions, and caste
- PC8.** empathize with the people with disability
- PC9.** offer support or help to a person with a disability only when asked
- PC10.** ensure to adhere to the guidelines laid in Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act
- PC11.** report any violation of prevention of sexual harassment (POSH) rules immediately to the POSH committee

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the guidelines laid on Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act
- KU2.** methods of workplace communication
- KU3.** methods of team coordination
- KU4.** the types of possible disabilities among people with disability (PWD)



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- KU5.** the challenges faced by PWD
- KU6.** importance of displaying empathy towards PWD
- KU7.** the right way to use the laws, acts, and provisions defined for PwD by the statutory bodies
- KU8.** the importance of awareness for gender sensitization and prevention of sexual harassment (POSH) act
- KU9.** the guidelines related to environmental sustainability
- KU10.** the WHO guidelines and ICH-cGMP rules for waste disposal and waste management

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to gauge the relevant information manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/comments
- GS2.** use written communication skills to record and communicate details of work done to appropriate stakeholders by using written or computer-based record/electronic mail
- GS3.** use verbal communication skills to communicate confidential and sensitive information discretely to the authorized person while interacting with teammates
- GS4.** use team-building skills while dealing with teammates to manage the difficult/stressful or emotional situations at work
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical thinking skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Environment sustainability</i>	15	25	5	5
PC1. ensure energy conservation by switching off the machine and equipment post operations	-	-	-	-
PC2. identify ways to optimize the usage of electricity/energy in various tasks/activities/processes	-	-	-	-
PC3. ensure energy conservation by optimizing the machine/ equipment performance	-	-	-	-
PC4. identify recyclable and non-recyclable, and hazardous waste generated	-	-	-	-
PC5. segregate waste into different categories to achieve minimum pollution of land and water	-	-	-	-
PC6. check for water leakage in plant/ work area and take corrective actions	-	-	-	-
<i>Sensitivity towards all genders and people with disability</i>	15	25	5	5
PC7. respect all genders, religions, and caste	-	-	-	-
PC8. empathize with the people with disability	-	-	-	-
PC9. offer support or help to a person with a disability only when asked	-	-	-	-
PC10. ensure to adhere to the guidelines laid in Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act	-	-	-	-
PC11. report any violation of prevention of sexual harassment (POSH) rules immediately to the POSH committee	-	-	-	-
NOS Total	30	50	10	10



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0119
NOS Name	Ensure environment sustainability and sensitivity towards all genders and people with disability
Sector	Life Sciences
Sub-Sector	Bio Pharmaceutical, Pharmaceutical, Contract Research
Occupation	Clinical Trials
NSQF Level	5.5
Credits	1.00
Version	3.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025

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LFS/N3506: Carry out management activities related for clinical trial

Description

This NOS is about an Associate-Clinical Research Management Individual Carrying out management activities related for clinical trial

Scope

The scope covers the following :

- Planning, Development, and Compliance
- Financial Management, Data Integrity, and Quality Assurance

Elements and Performance Criteria

Planning, Development, and Compliance

To be competent, the user/individual on the job must be able to:

- PC1.** Develop and implement comprehensive plans for the initiation, execution, and completion of clinical trials, ensuring alignment with project goals and timelines.
- PC2.** Monitor progress and make adjustments as necessary to meet milestones and deadlines.
- PC3.** Contribute to the creation and refinement of study protocols, ensuring they clearly define the methodology, objectives, and participant selection criteria.
- PC4.** Ensure the protocol aligns with scientific and regulatory standards, addressing all critical study requirements.
- PC5.** Identify and evaluate potential clinical trial sites based on patient demographics, facilities, and regulatory compliance.
- PC6.** Ensure site readiness and capability to meet trial requirements, including recruitment capacity and adherence to protocols.

Financial Management, Data Integrity, and Quality Assurance

To be competent, the user/individual on the job must be able to:

- PC7.** Manage the financial aspects of the clinical trial, ensuring adherence to the allocated budget.
- PC8.** Make informed decisions to optimize resource utilization while maintaining cost-effectiveness.
- PC9.** Oversee data collection and management processes to ensure accuracy, completeness, and compliance with regulatory standards.
- PC10.** Implement measures to prevent data discrepancies and ensure data quality throughout the trial.
- PC11.** Proactively identify and address challenges that may arise during the trial, ensuring patient safety and the integrity of the study.
- PC12.** Make timely and informed decisions to resolve issues effectively and maintain trial integrity.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** SOPs and organizational policies about communication, code of conduct
- KU2.** clinical research team reporting structure
- KU3.** correct method for carrying out corrective actions outlined for trial-related problems
- KU4.** sponsors and CRO roles and responsibilities
- KU5.** site roles and responsibilities
- KU6.** working in cross functional, cross geographical and cross -cultural teams
- KU7.** information security and confidentiality policy
- KU8.** clinical trial-related regulations and compliances
- KU9.** sample handling procedures
- KU10.** principles of trial protocol

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use written communication skills to accurately record the required details in a given format as per SOP and guidelines in the English language
- GS2.** use reading and comprehension skills of various coding systems to read instructions, guidelines, procedures, rules, and signage
- GS3.** use listening skills to interpret the instructions and procedures to be followed
- GS4.** use verbal communication skills to interact with supervisor, teammates, and cross-functional teams for coordination and to communicate confidential and sensitive information discretely to the authorized person
- GS5.** use team-building skills while dealing with teammates and while managing the difficult/stressful or emotional situations at work
- GS6.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS7.** apply planning and organizing skills to plan and organize tools and material required to fulfil documentation related requirements
- GS8.** apply critical thinking skills to analyze and identify what and when to report an issue/concern to the supervisor/ QA team / any other stakeholder
- GS9.** apply customer-centricity while generating and securing documents
- GS10.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS11.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Planning, Development, and Compliance</i>	15	30	5	10
PC1. Develop and implement comprehensive plans for the initiation, execution, and completion of clinical trials, ensuring alignment with project goals and timelines.	-	-	-	-
PC2. Monitor progress and make adjustments as necessary to meet milestones and deadlines.	-	-	-	-
PC3. Contribute to the creation and refinement of study protocols, ensuring they clearly define the methodology, objectives, and participant selection criteria.	-	-	-	-
PC4. Ensure the protocol aligns with scientific and regulatory standards, addressing all critical study requirements.	-	-	-	-
PC5. Identify and evaluate potential clinical trial sites based on patient demographics, facilities, and regulatory compliance.	-	-	-	-
PC6. Ensure site readiness and capability to meet trial requirements, including recruitment capacity and adherence to protocols.	-	-	-	-
<i>Financial Management, Data Integrity, and Quality Assurance</i>	10	20	5	5
PC7. Manage the financial aspects of the clinical trial, ensuring adherence to the allocated budget.	-	-	-	-
PC8. Make informed decisions to optimize resource utilization while maintaining cost-effectiveness.	-	-	-	-
PC9. Oversee data collection and management processes to ensure accuracy, completeness, and compliance with regulatory standards.	-	-	-	-
PC10. Implement measures to prevent data discrepancies and ensure data quality throughout the trial.	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Proactively identify and address challenges that may arise during the trial, ensuring patient safety and the integrity of the study.	-	-	-	-
PC12. Make timely and informed decisions to resolve issues effectively and maintain trial integrity.	-	-	-	-
NOS Total	25	50	10	15



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3506
NOS Name	Carry out management activities related for clinical trial
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials
NSQF Level	5.5
Credits	1.00
Version	1.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025



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DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services



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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

LFS/N3503: Perform site management activities at clinical research site

Description

This NOS unit is about Associate-Clinical Research Management (Pharma, Biologics and Medical devices) managing the activities at clinical research study site and communicating and coordinating with with colleagues in cross functional teams, principle investigator (PI) and other stakeholders

Scope

The scope covers the following :

- Manage clinical trial site activities
- Coordination and communication with team members
- Coordination with vendors and site staff

Elements and Performance Criteria

Manage clinical trial site activities

To be competent, the user/individual on the job must be able to:

- PC1.** observe study processes wherever necessary, to confirm adherence to study protocol and cGCP
- PC2.** review study documents and may observe study processes to confirm adherence to cGCP
- PC3.** schedule site initiation visits
- PC4.** review source documentation and data collection form
- PC5.** identify unreported events recorded in source documentation
- PC6.** perform IP counts for stock and returned supply against dispensing logs

Coordination and communication with team members

To be competent, the user/individual on the job must be able to:

- PC7.** work as a team with colleagues and share work as per their or own work load and skills
- PC8.** provide documented shift handovers to the next person in the shift or during transition from one project to other project
- PC9.** effectively communicate with team members in case of research related difficulties
- PC10.** escalate issues to manager or designated personnel in cases where support is required

Coordinating with vendors and site staff

To be competent, the user/individual on the job must be able to:

- PC11.** coordinate with the site/ CRO team for clinical research (in phase 1 or BA/ BE studies
- PC12.** coordinate with principal investigator, site manager, vendor, clinical research coordinator for clinical trial (in phase 2- phase 4 trial)
- PC13.** coordinate and follow up with site investigator, subject enrolment, trial performance, quality of study conduct, feedback during site monitoring, technical discussions about compliance to protocol
- PC14.** coordinate for internal or external quality audits, coordinating responses from the Principal Investigator(PI)



Qualification Pack

PC15. collect essential documents as per cGCP/ regulatory requirements, impart training to site staff

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** SOPs and organizational policies about communication, code of conduct
- KU2.** clinical research team reporting structure
- KU3.** correct method for carrying out corrective actions outlined for trial-related problems
- KU4.** sponsors' and CRO's roles and responsibilities
- KU5.** roles and responsibilities of every person at clinical research site
- KU6.** working in cross functional, cross geographical and cross -cultural teams
- KU7.** principles of clinical research study/ trial protocol

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use written communication skills to accurately record the required details in a given format as per SOP and guidelines in the English language
- GS2.** use reading and comprehension skills of various coding systems to read instructions, guidelines, procedures, rules, and signage
- GS3.** use listening skills to interpret the instructions and procedures to be followed
- GS4.** use verbal communication skills to interact with supervisor, teammates, and cross-functional teams for coordination and to communicate confidential and sensitive information discretely to the authorized person
- GS5.** use team-building skills while dealing with teammates and while managing the difficult/stressful or emotional situations at work
- GS6.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS7.** apply planning and organizing skills to plan and organize tools and material required to fulfill documentation related requirements
- GS8.** apply critical thinking skills to analyze and identify what and when to report an issue/concern to the supervisor/ QA team / any other stakeholder
- GS9.** apply customer-centricity while generating and securing documents
- GS10.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS11.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage clinical trial site activities</i>	10	20	5	5
PC1. observe study processes wherever necessary, to confirm adherence to study protocol and cGCP	-	-	-	-
PC2. review study documents and may observe study processes to confirm adherence to cGCP	-	-	-	-
PC3. schedule site initiation visits	-	-	-	-
PC4. review source documentation and data collection form	-	-	-	-
PC5. identify unreported events recorded in source documentation	-	-	-	-
PC6. perform IP counts for stock and returned supply against dispensing logs	-	-	-	-
<i>Coordination and communication with team members</i>	10	20	5	5
PC7. work as a team with colleagues and share work as per their or own work load and skills	-	-	-	-
PC8. provide documented shift handovers to the next person in the shift or during transition from one project to other project	-	-	-	-
PC9. effectively communicate with team members in case of research related difficulties	-	-	-	-
PC10. escalate issues to manager or designated personnel in cases where support is required	-	-	-	-
<i>Coordinating with vendors and site staff</i>	5	10	3	2
PC11. coordinate with the site/ CRO team for clinical research (in phase 1 or BA/ BE studies)	-	-	-	-
PC12. coordinate with principal investigator, site manager, vendor, clinical research coordinator for clinical trial (in phase 2- phase 4 trial)	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. coordinate and follow up with site investigator, subject enrolment, trial performance, quality of study conduct, feedback during site monitoring, technical discussions about compliance to protocol	-	-	-	-
PC14. coordinate for internal or external quality audits, coordinating responses from the Principal Investigator(PI)	-	-	-	-
PC15. collect essential documents as per cGCP/ regulatory requirements, impart training to site staff	-	-	-	-
NOS Total	25	50	13	12



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3503
NOS Name	Perform site management activities at clinical research site
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials, Bio Availability-Bio Equivalence
NSQF Level	5.5
Credits	7.0
Version	3.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025

Qualification Pack

LFS/N3504: Carry out reporting and documentation for site coordination activities as per regulatory standards

Description

This NOS is about Associate-Clinical Research Management performing the required reporting and documentation for site management activities as per protocol and regulatory standards

Scope

The scope covers the following :

- Recording and reporting for site management activities for Clinical Trial/ Bioavailability- Bioequivalence (BA-BE) Study
- Information security

Elements and Performance Criteria

Recording and reporting for site management activities for Clinical Trial/ Bioavailability- Bioequivalence (BA-BE) Study

To be competent, the user/individual on the job must be able to:

- PC1.** maintain documentation on clinical trial material shipping orders and prepare relevant reports
- PC2.** ensure Patient Information Sheet, Informed Consent Form, Patient Diaries, CRFs, development of regulatory dossier for regulatory and ethics committee submissions
- PC3.** ensure that adverse events are correctly documented and reported
- PC4.** assists the Principal Investigator in submission of accurate and timely closeout documents to applicable federal agencies
- PC5.** obtain filled documents/CRFs from the cross functional site teams
- PC6.** collect data as required by the protocol

Information security

To be competent, the user/individual on the job must be able to:

- PC7.** ensure that unfavorable occurrences (e.g. protocol deviations) are clearly reported and documented
- PC8.** coordinate with the pharmaco-vigilance teams for documenting post-marketing adverse drug reactions
- PC9.** respond to requests for information in an appropriate manner whilst following organizational procedures
- PC10.** make sure documents are available to all appropriate authorities to inspect/ audit

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** ICH-GCP guidelines, legislation and regulations as applicable and impact of non-conformance/poor practices
- KU2.** different methods of recording information
- KU3.** the reporting and recording formats
- KU4.** the importance of complete and accurate documentation
- KU5.** the Good Documentation Practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use written communication skills to accurately record the required details in a given format as per SOP and guidelines in the English language
- GS2.** use reading and comprehension skills of various coding systems to read instructions, guidelines, procedures, rules, and signage
- GS3.** use listening skills to interpret the instructions and procedures to be followed
- GS4.** use verbal communication skills to interact with supervisor, teammates, and cross-functional teams for coordination and to communicate confidential and sensitive information discretely to the authorized person
- GS5.** use team-building skills while dealing with teammates and while managing the difficult/stressful or emotional situations at work
- GS6.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS7.** apply planning and organizing skills to plan and organize tools and material required to fulfil documentation related requirements
- GS8.** apply critical thinking skills to analyze and identify what and when to report an issue/concern to the supervisor/ QA team / any other stakeholder
- GS9.** apply customer-centricity while generating and securing documents
- GS10.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS11.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Recording and reporting for site management activities for Clinical Trial/ Bioavailability-Bioequivalence (BA-BE) Study</i>	15	25	5	10
PC1. maintain documentation on clinical trial material shipping orders and prepare relevant reports	-	-	-	-
PC2. ensure Patient Information Sheet, Informed Consent Form, Patient Diaries, CRFs, development of regulatory dossier for regulatory and ethics committee submissions	-	-	-	-
PC3. ensure that adverse events are correctly documented and reported	-	-	-	-
PC4. assists the Principal Investigator in submission of accurate and timely closeout documents to applicable federal agencies	-	-	-	-
PC5. obtain filled documents/CRFs from the cross functional site teams	-	-	-	-
PC6. collect data as required by the protocol	-	-	-	-
<i>Information security</i>	10	20	5	10
PC7. ensure that unfavorable occurrences (e.g. protocol deviations) are clearly reported and documented	-	-	-	-
PC8. coordinate with the pharmaco-vigilance teams for documenting post-marketing adverse drug reactions	-	-	-	-
PC9. respond to requests for information in an appropriate manner whilst following organizational procedures	-	-	-	-
PC10. make sure documents are available to all appropriate authorities to inspect/ audit	-	-	-	-
NOS Total	25	45	10	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3504
NOS Name	Carry out reporting and documentation for site coordination activities as per regulatory standards
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials, Bio Availability-Bio Equivalence
NSQF Level	5.5
Credits	6.0
Version	3.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025

Qualification Pack

LFS/N3501: Monitor the clinical trial site to ensure that ICH GCP guidelines, study protocol and applicable regulations are followed

Description

This NOS is about Associate-Clinical Research Management performing clinical research study site monitoring activities to effectively support the clinical research project

Scope

The scope covers the following :

- Monitor clinical trials
- Monitor participants during a clinical trial for safety and rights of participants

Elements and Performance Criteria

Monitor clinical trials

To be competent, the user/individual on the job must be able to:

- PC1.** organize investigator's start-up meeting and study site initiation meetings
- PC2.** carry out clinical trials as per ICH GCP guidelines, study protocol, and applicable regulations
- PC3.** communicate effectively with principal investigator, co-investigator, and clinical research coordinators to gather relevant information
- PC4.** review and assess the clinical trial process
- PC5.** ensure to study procedures that are consistently carried out by the study team across all participants
- PC6.** ensure that principal investigator is submitting documents to the ethics committee in a timely manner
- PC7.** perform source data verification, review source documents, informed consent procedures and forms for evaluating the participant's eligibility and assessing the protection of participant's rights
- PC8.** review efficacy related aspects of the participants, investigational product compliance by participants, and review case report forms (CRFs)
- PC9.** conduct site audits to ensure that all investigational products are stored, and drug accountability is maintained as per SOP
- PC10.** ensure optimal usage of resources by effective deployment of the same

Monitor participants during a clinical trial for safety and rights of participants

To be competent, the user/individual on the job must be able to:

- PC11.** ensure safety and rights of participants during all phases of clinical trials
- PC12.** review safety events and ensure that drug-related Adverse Events (AE) are identified and promptly reported to all concerned stakeholders and ethics committee within prescribed timelines and as per SOPs
- PC13.** record and review the rate of subject recruitment, visits that subjects fail to make, and tests that are not conducted

Qualification Pack

- PC14.** ensure documentation exists at the site to follow up with the patient for any missing test/ procedure and recommend participant enrolment and retention plan with the principal investigator and co- investigator
- PC15.** ensure the documentation of the withdrawals of enrolled subjects with reasons on the CRFs by the site coordinators and principal investigators
- PC16.** identify anomalies in study conduct from a misconduct or fraud perspective

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organization SOPs
- KU2.** impact of various practices on cost, quality, productivity, delivery and safety
- KU3.** principles of clinical research
- KU4.** country regulations and compliances relevant to clinical trial process
- KU5.** physiology and reason of disease condition
- KU6.** standard of care, treatment options and dose of medication
- KU7.** clinical trial protocol
- KU8.** the characteristics of the investigational drug in the study
- KU9.** relevant licenses for study drug import and biological sample exports
- KU10.** procedures and responsibility for reporting research and performance information
- KU11.** basics of finance and accounts to keep a track of site payments and patient compensation
- KU12.** principles of ICH-GCP, Indian GCP and ICMR guidelines, Good Documentation Practices and Good Laboratory Practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read and understand manuals, SOPs, health and safety instructions, memos, reports, and notes/comments from the supervisor
- GS2.** use written communication skills to record and communicate details of work done to appropriate stakeholders by using written/typed report or computer-based record/electronic mail
- GS3.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS4.** apply critical thinking skills to analyze and identify when to report an issue/concern to the in-charge and when to deal with a colleague individually, depending on the type of concern
- GS5.** apply analytical skills in choosing a well-defined written smooth methods/instruction to resolve day to day problems
- GS6.** apply planning and organizing skills to plan and organize tools and material required to fulfil own work requirements on time

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Monitor clinical trials</i>	15	30	10	5
PC1. organize investigator's start-up meeting and study site initiation meetings	-	-	-	-
PC2. carry out clinical trials as per ICH GCP guidelines, study protocol, and applicable regulations	-	-	-	-
PC3. communicate effectively with principal investigator, co-investigator, and clinical research coordinators to gather relevant information	-	-	-	-
PC4. review and assess the clinical trial process	-	-	-	-
PC5. ensure to study procedures that are consistently carried out by the study team across all participants	-	-	-	-
PC6. ensure that principal investigator is submitting documents to the ethics committee in a timely manner	-	-	-	-
PC7. perform source data verification, review source documents, informed consent procedures and forms for evaluating the participant's eligibility and assessing the protection of participant's rights	-	-	-	-
PC8. review efficacy related aspects of the participants, investigational product compliance by participants, and review case report forms (CRFs)	-	-	-	-
PC9. conduct site audits to ensure that all investigational products are stored, and drug accountability is maintained as per SOP	-	-	-	-
PC10. ensure optimal usage of resources by effective deployment of the same	-	-	-	-
<i>Monitor participants during a clinical trial for safety and rights of participants</i>	10	20	5	5
PC11. ensure safety and rights of participants during all phases of clinical trials	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. review safety events and ensure that drug-related Adverse Events (AE) are identified and promptly reported to all concerned stakeholders and ethics committee within prescribed timelines and as per SOPs	-	-	-	-
PC13. record and review the rate of subject recruitment, visits that subjects fail to make, and tests that are not conducted	-	-	-	-
PC14. ensure documentation exists at the site to follow up with the patient for any missing test/ procedure and recommend participant enrolment and retention plan with the principal investigator and co- investigator	-	-	-	-
PC15. ensure the documentation of the withdrawals of enrolled subjects with reasons on the CRFs by the site coordinators and principal investigators	-	-	-	-
PC16. identify anomalies in study conduct from a misconduct or fraud perspective	-	-	-	-
NOS Total	25	50	15	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3501
NOS Name	Monitor the clinical trial site to ensure that ICH GCP guidelines, study protocol and applicable regulations are followed
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials, Bio Availability-Bio Equivalence
NSQF Level	5.5
Credits	7.0
Version	3.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025

Qualification Pack

LFS/N3502: Carry out reporting and documentation for site monitoring activities as per regulatory standards

Description

This NOS is about a Associate-Clinical Research Management performing the reporting and documentation for site monitoring activities

Scope

The scope covers the following :

- Pre research activities
- Research process activities
- Post research activities

Elements and Performance Criteria

Pre research activities

To be competent, the user/individual on the job must be able to:

- PC1.** outline the purpose and methodology of a trial
- PC2.** develop, draft, and write the Trial Protocols
- PC3.** present trial protocols to a steering committee
- PC4.** generate regulatory authority applications and approvals
- PC5.** identify, select and evaluate trial sites and investigators and provide inputs in investigator grants and agreements

Research process activities

To be competent, the user/individual on the job must be able to:

- PC6.** develop training materials for site for site initiations
- PC7.** prepare follow up letters to principal investigator, complete site monitoring documentation
- PC8.** prepare site monitoring visit reports, maintain records of communication with site staff and investigator, letters of agreement, lab reference ranges and schedule of payment
- PC9.** maintain project files including : ethics committee approvals; curriculum vitae of investigators and study personnel; Investigator's Undertaking etc.

Post research activities

To be competent, the user/individual on the job must be able to:

- PC10.** provide inputs to medical/ scientific teams as well as bio-statistician, who analyses technical trial data and writes technical trial reports
- PC11.** provide support in preparing final reports, occasionally manuscripts for publication
- PC12.** archive study documentation, and trial related correspondence

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** ICH-GCP guidelines, legislation and regulations as applicable and impact of non-conformance/poor practices
- KU2.** how to implement the relevant company SOPs for the fulfilment of each clinical trial
- KU3.** procedures and responsibility for reporting research and performance information
- KU4.** the reason and impact of the occurrence of problems
- KU5.** basics of finance and accounts to keep a track of site payments and patient compensation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills of various coding systems to read instructions, guidelines, procedures, rules, and signage
- GS2.** use listening skills to interpret the instructions and procedures to be followed
- GS3.** use verbal communication skills to interact with supervisor, teammates, and cross-functional teams for coordination and to communicate confidential and sensitive information discretely to the authorized person
- GS4.** use team-building skills while dealing with teammates and while managing the difficult/stressful or emotional situations at work
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply planning and organizing skills to plan and organize tools and material required to fulfil documentation related requirements
- GS7.** apply critical thinking skills to analyze and identify what and when to report an issue/concern to the supervisor/ QA team / any other stakeholder
- GS8.** apply customer-centricity while generating and securing documents
- GS9.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS10.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre research activities</i>	10	15	5	5
PC1. outline the purpose and methodology of a trial	-	-	-	-
PC2. develop, draft, and write the Trial Protocols	-	-	-	-
PC3. present trial protocols to a steering committee	-	-	-	-
PC4. generate regulatory authority applications and approvals	-	-	-	-
PC5. identify, select and evaluate trial sites and investigators and provide inputs in investigator grants and agreements	-	-	-	-
<i>Research process activities</i>	10	15	5	5
PC6. develop training materials for site for site initiations	-	-	-	-
PC7. prepare follow up letters to principal investigator, complete site monitoring documentation	-	-	-	-
PC8. prepare site monitoring visit reports, maintain records of communication with site staff and investigator, letters of agreement, lab reference ranges and schedule of payment	-	-	-	-
PC9. maintain project files including : ethics committee approvals; curriculum vitae of investigators and study personnel; Investigator's Undertaking etc.	-	-	-	-
<i>Post research activities</i>	5	15	5	5
PC10. provide inputs to medical/ scientific teams as well as bio-statistician, who analyses technical trial data and writes technical trial reports	-	-	-	-
PC11. provide support in preparing final reports, occasionally manuscripts for publication	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. archive study documentation, and trial related correspondence	-	-	-	-
NOS Total	25	45	15	15



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3502
NOS Name	Carry out reporting and documentation for site monitoring activities as per regulatory standards
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials
NSQF Level	5.5
Credits	6.0
Version	3.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025

Qualification Pack

LFS/N3505: Provide support for clinical data management activities

Description

This NOS is about an Associate-Clinical Research Management providing support for clinical data management activities

Scope

The scope covers the following :

- Designing eCRF
- Validation of clinical data

Elements and Performance Criteria

Designing eCRF

To be competent, the user/individual on the job must be able to:

- PC1.** assess database design requirements relevant to data capture tool, data extraction, processing, and reporting
- PC2.** eCRF design and edit check programming based on protocol requirements
- PC3.** manage the design specification, development, testing, and validation of electronic case report forms, edit checks standards, and dataset extraction
- PC4.** make recommendations to management concerning complex technical issues and provide solutions
- PC5.** manage multiple studies and projects simultaneously and deliver on the timelines
- PC6.** provide support in the fixing of any issues or changes to eCRF and edit checks encountered during system testing, User Acceptance Testing (UAT), and deployment testing
- PC7.** generate and maintain functional specifications for the eCRF edit checks
- PC8.** generate data listings to support data review
- PC9.** import external data in the EDC based on the data transfer agreement developed in collaboration with the clinical data manager
- PC10.** develop the eCRF completion guidelines

Validation of clinical data

To be competent, the user/individual on the job must be able to:

- PC11.** prepare and update annotation list, data validation plan, edit check specifications, and database/ data entry screens as per study requirements
- PC12.** perform query management activities per study requirements including the manual queries raised by the data entry team and medical coder
- PC13.** fill the data in 2 CRF(s) for data base testing
- PC14.** validate the external data after uploading it into the database
- PC15.** update self-evident correction queries in self-evident correction form per study requirements
- PC16.** update reports and trackers on regular basis or as per study requirements

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organization SOPs
- KU2.** impact of various practices on cost, quality, productivity, delivery, and safety
- KU3.** ICH-GCP guidelines, legislation and regulations as applicable and impact of non-conformance/poor practices
- KU4.** country regulations and compliances relevant to clinical trial process
- KU5.** clinical trial protocol
- KU6.** investigator brochure and knowledge of the characteristics of the investigational drug in the study
- KU7.** Good Documentation Practices and Medical Coding Dictionaries
- KU8.** clinical trial methodology
- KU9.** how to create eCRF specifications, managing eCRF timelines, designing and developing eCRFs
- KU10.** medical, clinical research, and data management process and terminology
- KU11.** data requirements and authoring specifications

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read and understand manuals, SOPs, health and safety instructions, memos, reports, and notes/comments
- GS2.** use written communication skills to record and communicate details of work done to appropriate stakeholders by using written/typed report or computer-based record/electronic mail
- GS3.** use written communication skills to maintain proper and concise records as per given format
- GS4.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS5.** apply critical thinking skills to analyze and identify when to report an issue/concern to the in-charge and when to deal with a colleague individually, depending on the type of concern
- GS6.** apply analytical skills in choosing a well-defined written smooth methods/instruction to resolve day to day problems
- GS7.** apply planning and organizing skills to plan and organize tools and material required to fulfil own work requirements on time
- GS8.** apply the analytical skill to observe investigations and identify deviations/ abnormal incidents
- GS9.** apply customer-centricity while generating and securing documents
- GS10.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS11.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Designing eCRF</i>	15	30	5	10
PC1. assess database design requirements relevant to data capture tool, data extraction, processing, and reporting	-	-	-	-
PC2. eCRF design and edit check programming based on protocol requirements	-	-	-	-
PC3. manage the design specification, development, testing, and validation of electronic case report forms, edit checks standards, and dataset extraction	-	-	-	-
PC4. make recommendations to management concerning complex technical issues and provide solutions	-	-	-	-
PC5. manage multiple studies and projects simultaneously and deliver on the timelines	-	-	-	-
PC6. provide support in the fixing of any issues or changes to eCRF and edit checks encountered during system testing, User Acceptance Testing (UAT), and deployment testing	-	-	-	-
PC7. generate and maintain functional specifications for the eCRF edit checks	-	-	-	-
PC8. generate data listings to support data review	-	-	-	-
PC9. import external data in the EDC based on the data transfer agreement developed in collaboration with the clinical data manager	-	-	-	-
PC10. develop the eCRF completion guidelines	-	-	-	-
<i>Validation of clinical data</i>	10	20	5	5
PC11. prepare and update annotation list, data validation plan, edit check specifications, and database/ data entry screens as per study requirements	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. perform query management activities per study requirements including the manual queries raised by the data entry team and medical coder	-	-	-	-
PC13. fill the data in 2 CRF(s) for data base testing	-	-	-	-
PC14. validate the external data after uploading it into the database	-	-	-	-
PC15. update self-evident correction queries in self-evident correction form per study requirements	-	-	-	-
PC16. update reports and trackers on regular basis or as per study requirements	-	-	-	-
NOS Total	25	50	10	15



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3505
NOS Name	Provide support for clinical data management activities
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials
NSQF Level	5.5
Credits	13.0
Version	3.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025

Qualification Pack

LFS/N3508: Provide support in data through Biostatistical analysis

Description

This NOS is about a Associate-Clinical Research Management providing support in data through Biostatistical analysis

Scope

The scope covers the following :

- Data Preparation and Refinement for Statistical Analysis
- Data Visualization for Statistical Analysis

Elements and Performance Criteria

Data Preparation and Refinement for Statistical Analysis

To be competent, the user/individual on the job must be able to:

- PC1.** define the format, structure, and statistical requirements of the research/ clinical dataset to ensure compatibility with analytical methods.
- PC2.** establish indexing strategies and organize variables systematically for efficient statistical computation
- PC3.** identify and classify data types (continuous, categorical, ordinal, etc.) for each variable to ensure appropriate statistical treatment.
- PC4.** detect and impute missing values using statistical techniques such as mean/mode imputation, regression, or multiple imputation methods.
- PC5.** identify and correct incorrect data types to prevent analytical errors in statistical modelling.
- PC6.** sort and segment data into meaningful subsets to facilitate hypothesis testing and subgroup analysis.
- PC7.** perform necessary transformations (e.g., normalization, standardization, log transformation) to meet statistical assumptions.
- PC8.** identify redundant data points and apply data normalization techniques to reduce bias and improve model accuracy.
- PC9.** validate pre-processed data using statistical diagnostics, such as descriptive statistics and outlier detection methods.
- PC10.** ensure compliance with data privacy regulations (e.g., HIPAA, GDPR) while maintaining data integrity for analysis.

Data Visualization for Statistical Analysis

To be competent, the user/individual on the job must be able to:

- PC11.** identify statistical objectives, such as trend analysis, hypothesis testing, or predictive modelling, to guide data visualization.
- PC12.** define the purpose, scope, and target audience for statistical reporting of results
- PC13.** determine the optimal delivery mode and format (e.g., interactive dashboards, statistical reports, apis) for conveying insights.

Qualification Pack

- PC14.** summarize key statistical findings from the research/ clinical dataset into a structured and meaningful narrative.
- PC15.** select appropriate visualization techniques (e.g., box plots, histograms, kaplan-meier survival curves, forest plots) to represent statistical results effectively.
- PC16.** present statistical outputs using standardized templates and consistent language for clarity and reproducibility.
- PC17.** validate visualizations by cross-referencing with statistical outputs and stakeholder feedback.
- PC18.** publish validated visualizations in agreed formats for dissemination across research, regulatory, and industry platforms.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** ICH-GCP guidelines, legislation and regulations as applicable and impact of non-conformance/poor practices
- KU2.** basics of Object-Oriented Programming like (C plus plus / JAVA), JavaScript, R and Python / Perl and operating system like Linux .
- KU3.** factors which may affect product output and must be aware of appropriate remedies
- KU4.** basic Statistical concepts and machine learning concepts
- KU5.** use of distributed computing systems
- KU6.** various image formats for analysis
- KU7.** different data sources and how to access documents and information from data sources
- KU8.** organizational policies and procedures for sharing data
- KU9.** different statistical concepts such as distributions, hypothesis testing, confidence intervals etc.
- KU10.** different graphical formats for presenting data and how to create these
- KU11.** different styles used in visualizations, including your organizations house style, types & templates
- KU12.** different applications, libraries or packages to create visualizations using tools such as Tableau, Qlikview, etc
- KU13.** how to work on various operating systems such as linux, ubuntu, or windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read and understand manuals, SOPs, health and safety instructions, memos, reports, and notes/comments
- GS2.** use written communication skills to record and communicate details of work done to appropriate stakeholders by using written/typed report or computer-based record/electronic mail
- GS3.** use written communication skills to maintain proper and concise records as per given format
- GS4.** apply problem-solving skills to find solutions for workflow-related difficulties



Qualification Pack

- GS5.** apply critical thinking skills to analyze and identify when to report an issue/concern to the in-charge and when to deal with a colleague individually, depending on the type of concern
- GS6.** apply analytical skills in choosing a well-defined written smooth methods/instruction to resolve day to day problems
- GS7.** apply planning and organizing skills to plan and organize tools and material required to fulfil own work requirements on time
- GS8.** apply the analytical skill to observe investigations and identify deviations/ abnormal incidents
- GS9.** apply customer-centricity while generating and securing documents
- GS10.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS11.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations
- GS12.** ability and willingness to adjust quickly to new situations in a continuously developing environment

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Data Preparation and Refinement for Statistical Analysis</i>	15	20	5	10
PC1. define the format, structure, and statistical requirements of the research/ clinical dataset to ensure compatibility with analytical methods.	-	-	-	-
PC2. establish indexing strategies and organize variables systematically for efficient statistical computation	-	-	-	-
PC3. identify and classify data types (continuous, categorical, ordinal, etc.) for each variable to ensure appropriate statistical treatment.	-	-	-	-
PC4. detect and impute missing values using statistical techniques such as mean/mode imputation, regression, or multiple imputation methods.	-	-	-	-
PC5. identify and correct incorrect data types to prevent analytical errors in statistical modelling.	-	-	-	-
PC6. sort and segment data into meaningful subsets to facilitate hypothesis testing and subgroup analysis.	-	-	-	-
PC7. perform necessary transformations (e.g., normalization, standardization, log transformation) to meet statistical assumptions.	-	-	-	-
PC8. identify redundant data points and apply data normalization techniques to reduce bias and improve model accuracy.	-	-	-	-
PC9. validate pre-processed data using statistical diagnostics, such as descriptive statistics and outlier detection methods.	-	-	-	-
PC10. ensure compliance with data privacy regulations (e.g., HIPAA, GDPR) while maintaining data integrity for analysis.	-	-	-	-
<i>Data Visualization for Statistical Analysis</i>	20	20	5	5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. identify statistical objectives, such as trend analysis, hypothesis testing, or predictive modelling, to guide data visualization.	-	-	-	-
PC12. define the purpose, scope, and target audience for statistical reporting of results	-	-	-	-
PC13. determine the optimal delivery mode and format (e.g., interactive dashboards, statistical reports, apis) for conveying insights.	-	-	-	-
PC14. summarize key statistical findings from the research/ clinical dataset into a structured and meaningful narrative.	-	-	-	-
PC15. select appropriate visualization techniques (e.g., box plots, histograms, kaplan-meier survival curves, forest plots) to represent statistical results effectively.	-	-	-	-
PC16. present statistical outputs using standardized templates and consistent language for clarity and reproducibility.	-	-	-	-
PC17. validate visualizations by cross-referencing with statistical outputs and stakeholder feedback.	-	-	-	-
PC18. publish validated visualizations in agreed formats for dissemination across research, regulatory, and industry platforms.	-	-	-	-
NOS Total	35	40	10	15



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3508
NOS Name	Provide support in data through Biostatistical analysis
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials
NSQF Level	5.5
Credits	4.0
Version	1.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025

Qualification Pack

LFS/N3509: Provide support in PK & PD through Biostatistics analysis

Description

This NOS is about a Associate-Clinical Research Management providing support in PK & PD through Biostatistics analysis

Scope

The scope covers the following :

- Support in PK & PD studies
- Support in Data Management, Imputation, and Reporting Tools

Elements and Performance Criteria

Support in PK & PD studies

To be competent, the user/individual on the job must be able to:

- PC1.** ensure appropriate design with unbiased randomization, allocation concealment, and blinding.
- PC2.** justify sample size and effect size based on literature, ensuring statistical and clinical relevance.
- PC3.** provide statistical input in planning interim analyses with clear stopping rules for safety and efficacy oversight
- PC4.** provide statistical expertise for manuscripts, and regulatory documents with accurate and consistent methodologies.
- PC5.** provide statistical consulting to ensure compliance with regulatory requirements for analysis and interpretation.

Support in Data Management, Imputation, and Reporting Tools

To be competent, the user/individual on the job must be able to:

- PC6.** ensure report and diaries capture necessary data clearly and in line with study objectives
- PC7.** perform calculations such as Cmax and AUC to analyze pharmacodynamic and pharmacokinetic data.
- PC8.** develop a Statistical Analysis Plan (SAP) with justified data imputation methods and handling of missing data.
- PC9.** utilize various software tools like Winclin and Phoenix for statistical analysis.
- PC10.** statistical models should be appropriate for the study design and objectives, and their assumptions should be checked and justified.
- PC11.** ensure accurate Tables, listings, and figures should be clear, concise, and properly formatted.
- PC12.** ensure that TLF generation is automated or done via reproducible programming practices to reduce error risk.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** ICH-GCP guidelines, legislation and regulations as applicable and impact of non-conformance/poor practices
- KU2.** the organization's policies, procedures and guidelines which relate to rule-based statistical analysis
- KU3.** impact of various practices on cost, quality, productivity, delivery and safety
- KU4.** different sources of data and accessing information from data sources complying with organizational policy of data sharing
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** different methodological approaches to statistical analysis
- KU7.** techniques for interrogating and validating statistical data
- KU8.** different databases of statistical analysis, software and tools and how to use these, including SQL and scripting tools

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read and understand manuals, SOPs, health and safety instructions, memos, reports, and notes/comments
- GS2.** use written communication skills to record and communicate details of work done to appropriate stakeholders by using written/typed report or computer-based record/electronic mail
- GS3.** use written communication skills to maintain proper and concise records as per given format
- GS4.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS5.** apply critical thinking skills to analyze and identify when to report an issue/concern to the in-charge and when to deal with a colleague individually, depending on the type of concern
- GS6.** apply analytical skills in choosing a well-defined written smooth methods/instruction to resolve day to day problems
- GS7.** apply planning and organizing skills to plan and organize tools and material required to fulfil own work requirements on time
- GS8.** apply the analytical skill to observe investigations and identify deviations/ abnormal incidents
- GS9.** apply customer-centricity while generating and securing documents
- GS10.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS11.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations
- GS12.** ability and willingness to adjust quickly to new situations in a continuously developing environment

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Support in PK & PD studies</i>	15	30	5	10
PC1. ensure appropriate design with unbiased randomization, allocation concealment, and blinding.	-	-	-	-
PC2. justify sample size and effect size based on literature, ensuring statistical and clinical relevance.	-	-	-	-
PC3. provide statistical input in planning interim analyses with clear stopping rules for safety and efficacy oversight	-	-	-	-
PC4. provide statistical expertise for manuscripts, and regulatory documents with accurate and consistent methodologies.	-	-	-	-
PC5. provide statistical consulting to ensure compliance with regulatory requirements for analysis and interpretation.	-	-	-	-
<i>Support in Data Management, Imputation, and Reporting Tools</i>	10	20	5	5
PC6. ensure report and diaries capture necessary data clearly and in line with study objectives	-	-	-	-
PC7. perform calculations such as Cmax and AUC to analyze pharmacodynamic and pharmacokinetic data.	-	-	-	-
PC8. develop a Statistical Analysis Plan (SAP) with justified data imputation methods and handling of missing data.	-	-	-	-
PC9. utilize various software tools like Vinolin and Phoenix for statistical analysis.	-	-	-	-
PC10. statistical models should be appropriate for the study design and objectives, and their assumptions should be checked and justified.	-	-	-	-
PC11. ensure accurate Tables, listings, and figures should be clear, concise, and properly formatted.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure that TLF generation is automated or done via reproducible programming practices to reduce error risk.	-	-	-	-
NOS Total	25	50	10	15



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3509
NOS Name	Provide support in PK & PD through Biostatistics analysis
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials
NSQF Level	5.5
Credits	4.0
Version	1.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025

Qualification Pack

LFS/N3510: Provide algorithm support for Biostatistics activities

Description

This NOS is about a Associate-Clinical Research Management providing algorithm support for Biostatistics activities

Scope

The scope covers the following :

- Algorithm Creation for Statistical Analysis
- Statistical Analysis using available / customized algorithm

Elements and Performance Criteria

Algorithm Creation for Statistical Analysis

To be competent, the user/individual on the job must be able to:

- PC1.** define the mathematical foundation and statistical principles for developing algorithms tailored to research/clinical data analysis.
- PC2.** design and prototype new statistical algorithms to handle research/clinical -specific challenges, such as adaptive research designs, missing data imputation, and longitudinal data analysis.
- PC3.** develop machine learning and artificial intelligence-based models to enhance predictive analytics, patient stratification, and biomarker discovery in research/clinical.
- PC4.** optimize algorithm efficiency by implementing computational techniques such as parallel processing, cloud computing, and memory-efficient data handling.
- PC5.** Validate algorithm performance through benchmarking against existing statistical models and research/clinical datasets, ensuring accuracy, reproducibility, and regulatory compliance.

Statistical Analysis using available and customized algorithm

To be competent, the user/individual on the job must be able to:

- PC6.** refine and implement statistical algorithms, programming codes, and analytical methodologies to manage, process, and analyze research/clinical data effectively.
- PC7.** apply statistical programming languages (e.g., SAS, R, Python) and mathematical modeling techniques to design and implement methods for analyzing research/clinical endpoints, patient demographics, and treatment outcomes.
- PC8.** utilize biostatistical methods such as survival analysis, mixed-effects models, and propensity score matching to derive meaningful research /clinical insights.
- PC9.** perform quality control and sensitivity analysis to validate the robustness of statistical models used in research/clinical.
- PC10.** communicate algorithmic refinements, customizations, and validation results to the development team and regulatory stakeholders to ensure compliance with research/clinical protocols and guidelines (e.g., ICH E9, FDA, EMA).

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** ICH-GCP guidelines, legislation and regulations as applicable and impact of non-conformance/poor practices
- KU2.** the organization's policies, procedures and guidelines which relate to rule-based statistical analysis
- KU3.** different sources of data and accessing information from data sources complying with organizational policy of data sharing
- KU4.** the range of standard templates and tools available and how to use them
- KU5.** different methodological approaches to statistical analysis
- KU6.** techniques for interrogating and validating statistical data
- KU7.** different databases of statistical analysis, software and tools and how to use these, including SQL and scripting tools

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read and understand manuals, SOPs, health and safety instructions, memos, reports, and notes/comments
- GS2.** use written communication skills to record and communicate details of work done to appropriate stakeholders by using written/typed report or computer-based record/electronic mail
- GS3.** use written communication skills to maintain proper and concise records as per given format
- GS4.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS5.** apply critical thinking skills to analyze and identify when to report an issue/concern to the in-charge and when to deal with a colleague individually, depending on the type of concern
- GS6.** apply analytical skills in choosing a well-defined written smooth methods/instruction to resolve day to day problems
- GS7.** apply planning and organizing skills to plan and organize tools and material required to fulfil own work requirements on time
- GS8.** apply the analytical skill to observe investigations and identify deviations/ abnormal incidents
- GS9.** apply customer-centricity while generating and securing documents
- GS10.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS11.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations
- GS12.** ability and willingness to adjust quickly to new situations in a continuously developing environment

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Algorithm Creation for Statistical Analysis</i>	15	30	5	10
PC1. define the mathematical foundation and statistical principles for developing algorithms tailored to research/clinical data analysis.	-	-	-	-
PC2. design and prototype new statistical algorithms to handle research/clinical -specific challenges, such as adaptive research designs, missing data imputation, and longitudinal data analysis.	-	-	-	-
PC3. develop machine learning and artificial intelligence-based models to enhance predictive analytics, patient stratification, and biomarker discovery in research/clinical.	-	-	-	-
PC4. optimize algorithm efficiency by implementing computational techniques such as parallel processing, cloud computing, and memory-efficient data handling.	-	-	-	-
PC5. Validate algorithm performance through benchmarking against existing statistical models and research/clinical datasets, ensuring accuracy, reproducibility, and regulatory compliance.	-	-	-	-
<i>Statistical Analysis using available and customized algorithm</i>	10	20	5	5
PC6. refine and implement statistical algorithms, programming codes, and analytical methodologies to manage, process, and analyze research/clinical data effectively.	-	-	-	-
PC7. apply statistical programming languages (e.g., SAS, R, Python) and mathematical modeling techniques to design and implement methods for analyzing research/clinical endpoints, patient demographics, and treatment outcomes.	-	-	-	-
PC8. utilize biostatistical methods such as survival analysis, mixed-effects models, and propensity score matching to derive meaningful research /clinical insights.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. perform quality control and sensitivity analysis to validate the robustness of statistical models used in research/clinical.	-	-	-	-
PC10. communicate algorithmic refinements, customizations, and validation results to the development team and regulatory stakeholders to ensure compliance with research/clinical protocols and guidelines (e.g., ICH E9, FDA, EMA).	-	-	-	-
NOS Total	25	50	10	15



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3510
NOS Name	Provide algorithm support for Biostatistics activities
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials
NSQF Level	5.5
Credits	4.0
Version	1.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025



Qualification Pack

LFS/N3511: Apply AI/ML Models in biostatistical analysis

Description

This NOS is about a Associate-Clinical Research Management applying AI/ML Models in biostatistical analysis

Scope

The scope covers the following :

- Apply AI/ML Models

Elements and Performance Criteria

Apply AI/ML Models

To be competent, the user/individual on the job must be able to:

- PC1.** describe various ways AI/ML improves predictions in clinical research.
- PC2.** explain ML-based stratification techniques with relevant applications.
- PC3.** identify and explain various clustering techniques with correct mention of data types.
- PC4.** outline the process and ML techniques used in biomarker discovery.
- PC5.** list and describe various algorithms with mention of their predictive strengths and use cases.
- PC6.** list ethical concerns and key regulatory frameworks with examples.
- PC7.** explain various features that improve model performance
- PC8.** utilize explainable AI tools (e.g., SHAP, LIME) to interpret predictions and ensure transparency.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** role of AI/ML models in enhancing predictive analytics
- KU2.** ICH-GCP guidelines, legislation and regulations as applicable and impact of non-conformance/poor practices
- KU3.** the organization's policies, procedures and guidelines which relate to rule-based statistical analysis
- KU4.** different sources of data and accessing information from data sources complying with organizational policy of data sharing
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** different methodological approaches to statistical analysis
- KU7.** techniques for interrogating and validating statistical data
- KU8.** different databases of statistical analysis, software and tools and how to use these, including SQL and scripting tools
- KU9.** ethical, regulatory, and data privacy considerations when applying AI/ML models in life sciences



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read and understand manuals, SOPs, health and safety instructions, memos, reports, and notes/comments
- GS2.** use written communication skills to record and communicate details of work done to appropriate stakeholders by using written/typed report or computer-based record/electronic mail
- GS3.** use written communication skills to maintain proper and concise records as per given format
- GS4.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS5.** apply critical thinking skills to analyze and identify when to report an issue/concern to the in-charge and when to deal with a colleague individually, depending on the type of concern
- GS6.** apply analytical skills in choosing a well-defined written smooth methods/instruction to resolve day to day problems
- GS7.** apply planning and organizing skills to plan and organize tools and material required to fulfil own work requirements on time
- GS8.** apply the analytical skill to observe investigations and identify deviations/ abnormal incidents
- GS9.** apply customer-centricity while generating and securing documents
- GS10.** apply customer-centricity to remain compliant with data integrity rules, GMP guidelines and to evaluate the impact of wrongdoings
- GS11.** apply decision-making skills to make balanced judgments within the authority while dealing with daily work-life situations
- GS12.** ability and willingness to adjust quickly to new situations in a continuously developing environment

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Apply AI/ML Models</i>	25	50	10	15
PC1. describe various ways AI/ML improves predictions in clinical research.	-	-	-	-
PC2. explain ML-based stratification techniques with relevant applications.	-	-	-	-
PC3. identify and explain various clustering techniques with correct mention of data types.	-	-	-	-
PC4. outline the process and ML techniques used in biomarker discovery.	-	-	-	-
PC5. list and describe various algorithms with mention of their predictive strengths and use cases.	-	-	-	-
PC6. list ethical concerns and key regulatory frameworks with examples.	-	-	-	-
PC7. explain various features that improve model performance	-	-	-	-
PC8. utilize explainable AI tools (e.g., SHAP, LIME) to interpret predictions and ensure transparency.	-	-	-	-
NOS Total	25	50	10	15

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N3511
NOS Name	Apply AI/ML Models in biostatistical analysis
Sector	Life Sciences
Sub-Sector	Contract Research
Occupation	Clinical Trials
NSQF Level	5.5
Credits	1.0
Version	1.0
Last Reviewed Date	08/04/2025
Next Review Date	08/04/2028
NSQC Clearance Date	08/04/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by Life Sciences Sector Skill Development Council (LSSSDC)
2. Each Element will be assigned marks proportional to its importance in NOS. LSSSDC will also lay down the proportion of marks for Theory, Practical, Project, and Viva for each Element.
3. The assessment for the theory part will be based on the knowledge bank of questions created by the LSSSDC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. LSSSDC as assessment and awarding body will create unique evaluations for each assessment component i.e. theory, practical, project and viva for every student at each examination/training center based on this criterion.
6. Wherever any assessment component is not applicable/ feasible, the balance assessment components will be used to assess the candidate and accordingly the total marks will be calculated only for the applied



Qualification Pack

assessment component.

7. To pass the Qualification Pack, every trainee should score a minimum of 50-70%% of marks in each NOS to successfully clear the assessment. In the case of a Govt funded program, the program guidelines will be overarching on the pass percentage rules.

8. In case of unsuccessful completion, the trainee may seek re-assessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3507.Introduction to life sciences industry and basics of Clinical trial occupation	60	0	20	20	100	10
LFS/N0119.Ensure environment sustainability and sensitivity towards all genders and people with disability	30	50	10	10	100	10
LFS/N3506.Carry out management activities related for clinical trial	25	50	10	15	100	10
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
Total	135	130	40	45	350	40

Elective: 1 Site management

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3503.Perform site management activities at clinical research site	25	50	13	12	100	30
LFS/N3504.Carry out reporting and documentation for site coordination activities as per regulatory standards	25	45	10	20	100	30
Total	50	95	23	32	200	60

Elective: 2 Study Monitoring

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3501.Monitor the clinical trial site to ensure that ICH GCP guidelines, study protocol and applicable regulations are followed	25	50	15	10	100	30
LFS/N3502.Carry out reporting and documentation for site monitoring activities as per regulatory standards	25	45	15	15	100	30
Total	50	95	30	25	200	60

Elective: 3 Data management

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3505.Provide support for clinical data management activities	25	50	10	15	100	60



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	25	50	10	15	100	60

Elective: 4 Biostatistics

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N3508.Provide support in data through Biostatistical analysis	35	40	10	15	100	20
LFS/N3509.Provide support in PK & PD through Biostatistics analysis	25	50	10	15	100	20
LFS/N3510.Provide algorithm support for Biostatistics activities	25	50	10	15	100	10
LFS/N3511.Apply AI/ML Models in biostatistical analysis	25	50	10	15	100	10
Total	110	190	40	60	400	60

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
POSH	Prevention Of Sexual Harassment
PWD	People With Disability
SOP	Standard Operating Procedure
CRO	Clinical Research Organization
GCP	Good Clinical Practices
GDP	Good Documentation Practices
ICH	International Council for Harmonisation
GCP	Good Clinical Practices
CRF	Case Report Forms
AE	Adverse Events
SOP	Standard Operating Procedures
SOP	Standard Operating Procedures
GDP	Good Documentation Practices
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GDP	Good Documentation Practices
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SOP	Standard Operating Procedures
GDP	Good Documentation Practices
ICH	International Council for Harmonisation
GCP	Good Clinical Practices
SOP	Standard Operating Procedures
HIPAA	Health Insurance Portability and Accountability Act
GDPR	General Data Protection Regulation
SOP	Standard Operating Procedures
HIPAA	Health Insurance Portability and Accountability Act
GDPR	General Data Protection Regulation
SAP	Statistical Analysis Plan
SOP	Standard Operating Procedures



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HIPAA	Health Insurance Portability and Accountability Act
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HIPAA	Health Insurance Portability and Accountability Act
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GDPR	General Data Protection Regulation
SAP	Statistical Analysis Plan

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.